

Microsoft Sentinel Threat Hunting and Detection Engineering Lab

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Executive Summary

This project demonstrates the use of Microsoft Sentinel and Microsoft Defender XDR to perform threat hunting, telemetry analysis, and detection engineering activities within a Security Operations Center (SOC) environment. A Microsoft Sentinel workspace was integrated with Microsoft Defender XDR to collect endpoint telemetry and provide centralized visibility into security events. Advanced Hunting was utilized to analyze process execution, network communications, authentication activity, and registry modifications. Threat hunting activities focused on identifying PowerShell execution, reconnaissance activity, and persistence-related behavior. Finally, custom detection rules were developed to automate monitoring and alert generation for suspicious activity. This project demonstrates practical experience with SIEM operations, threat hunting, detection engineering, and MITRE ATT&CK-aligned investigations.

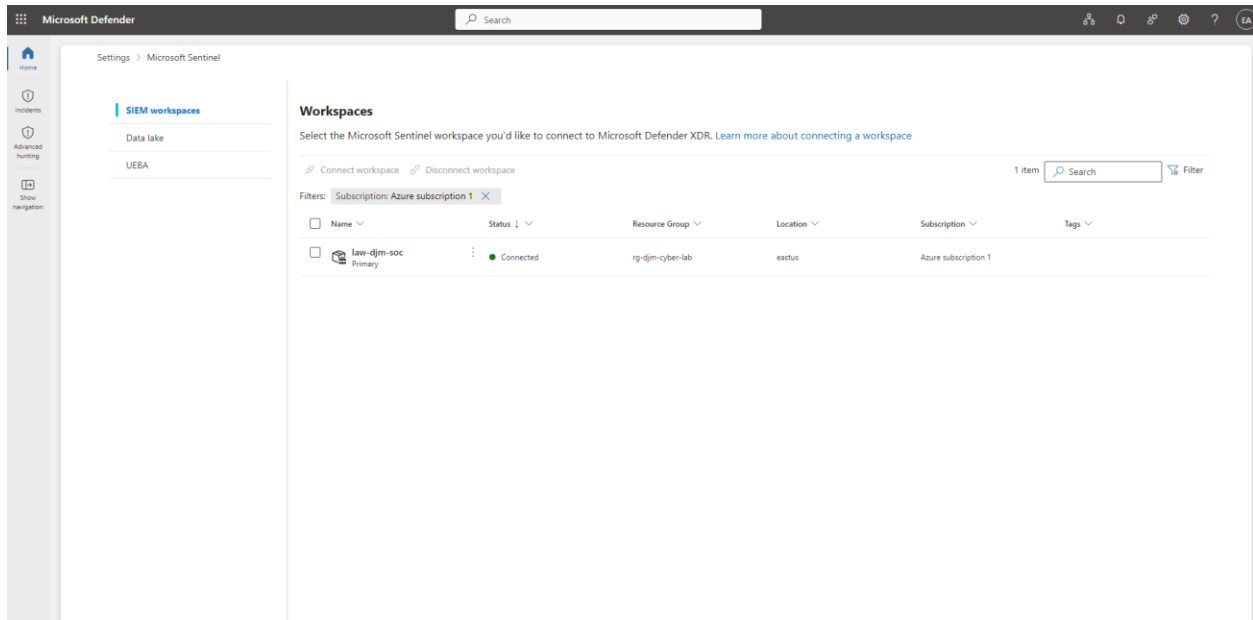
Lab Environment

The lab environment consisted of Microsoft Sentinel integrated with Microsoft Defender XDR and Microsoft Defender for Endpoint. The Windows 11 endpoint was onboarded to Microsoft Defender for Endpoint, enabling telemetry collection and advanced hunting capabilities.

Components

- Microsoft Sentinel
- Microsoft Defender XDR
- Microsoft Defender for Endpoint (MDE)
- Windows 11 Endpoint (djm)
- Azure Subscription
- Log Analytics Workspace

Figure 1 – Microsoft Sentinel Workspace Connected to Microsoft Defender XDR



Microsoft Sentinel and Defender XDR Configuration

Microsoft Sentinel was configured and connected to Microsoft Defender XDR to provide centralized security monitoring and investigation capabilities. The integration enabled incident correlation, threat visibility, and advanced hunting across endpoint telemetry sources.

Figure 2 – Microsoft Sentinel Incident Queue Showing Active Security Incident

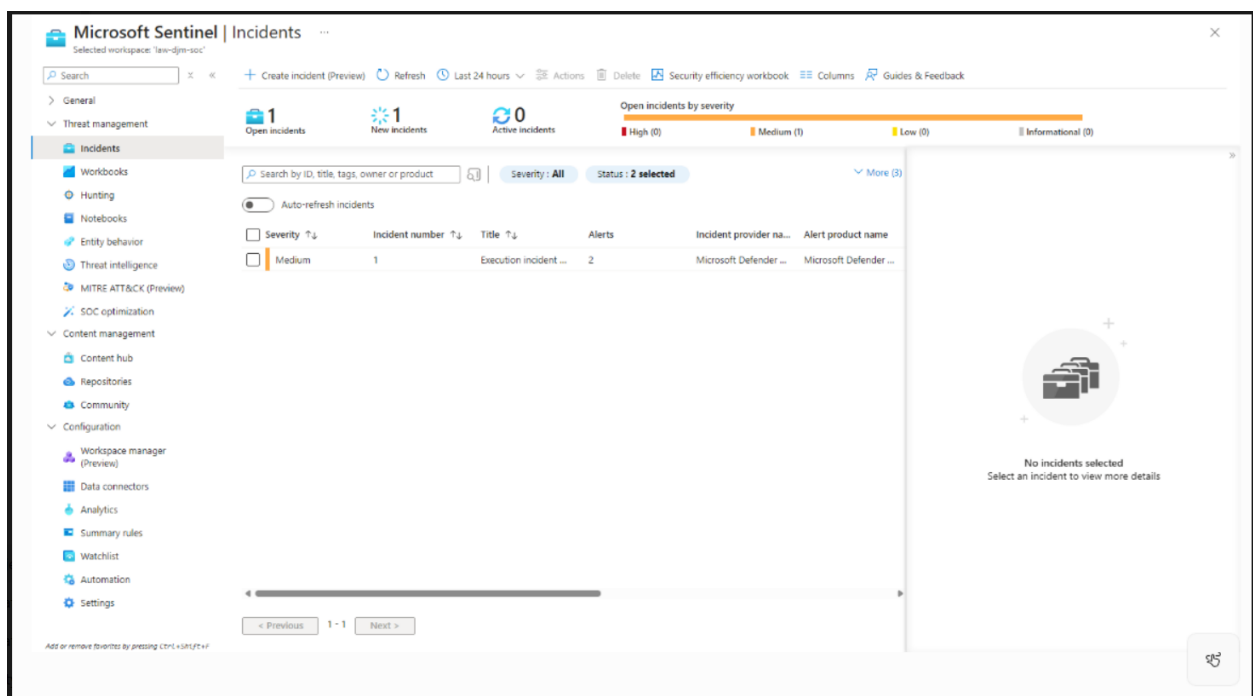
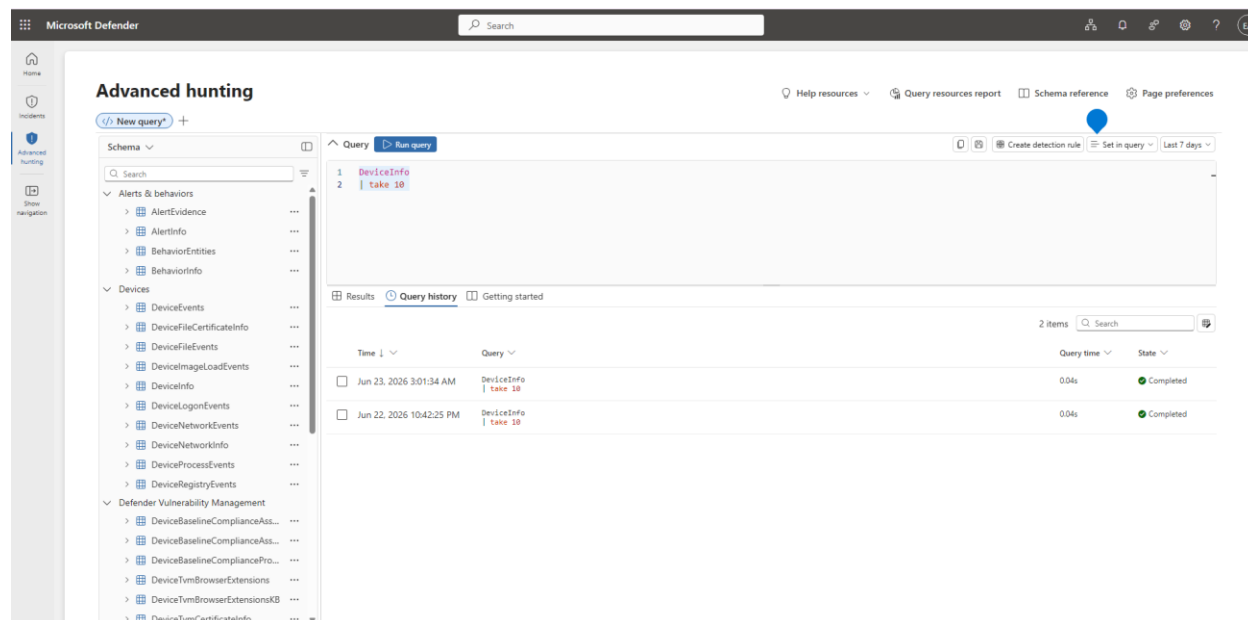


Figure 3 – Microsoft Defender Advanced Hunting Workspace



Telemetry Validation

Prior to conducting threat hunting activities, telemetry sources were validated to ensure endpoint data was being collected and made available for analysis.

Endpoint Telemetry Verification

The DeviceInfo table was queried to verify successful endpoint onboarding and telemetry collection.

Figure 4 – Endpoint Telemetry Verification Using the DeviceInfo Table

Advanced hunting

Help resources Query resources report Schema reference Page preferences

New query* +

Schema

Alerts & behaviors

- AlertEvidence
- AlertInfo
- BehaviorEntities
- BehaviorInfo

Devices

- DeviceEvents
- DeviceFileCertificateInfo
- DeviceFileEvents
- DeviceImageLoadEvents
- DeviceInfo
- DeviceLogonEvents
- DeviceNetworkEvents
- DeviceNetworkInfo
- DeviceProcessEvents
- DeviceRegistryEvents
- Defender Vulnerability Management
- DeviceBaselineComplianceAss...
- DeviceBaselineComplianceAss...
- DeviceBaselineCompliancePro...
- DeviceTvmBrowserExtensions
- DeviceTvmBrowserExtensionsKB
- DeviceTvmCertificateInfo

Query

1 DeviceInfo

2 take 18

Results Query history Getting started

10 items Search 00:02:327 Low

Filters: Add filter

TimeGenerated	Timestamp	DeviceId	DeviceName	ClientVersion	PublicIP	OSArchitecture	OSPlatform
Jun 23, 2026 2:05:...	Jun 23, 2026 2:05:02 AM	4d95e187d0045b63ff...	djm	10.8830.26200.8655	35.147.147.52	64-bit	Windows11
Jun 23, 2026 2:05:...	Jun 23, 2026 2:05:02 AM	4d95e187d0045b63ff...	djm	10.8830.26200.8655	35.147.147.52	64-bit	Windows11
Jun 23, 2026 3:35:...	Jun 23, 2026 3:35:08 AM	4d95e187d0045b63ff...	djm	10.8830.26200.8655	35.147.147.52	64-bit	Windows11
Jun 23, 2026 3:35:...	Jun 23, 2026 3:35:08 AM	4d95e187d0045b63ff...	djm	10.8830.26200.8655	35.147.147.52	64-bit	Windows11
Jun 23, 2026 4:18:...	Jun 23, 2026 4:18:24 AM	4d95e187d0045b63ff...	djm	10.8830.26200.8655	35.147.147.52	64-bit	Windows11
Jun 23, 2026 2:33:...	Jun 23, 2026 2:33:23 AM	4d95e187d0045b63ff...	djm	10.8830.26200.8655	35.147.147.52	64-bit	Windows11
Jun 22, 2026 11:05:...	Jun 22, 2026 11:05:24 PM	1b397c27d7d9bd0ef...	djm	1.1500		64-bit	Windows11
Jun 23, 2026 2:02:...	Jun 23, 2026 2:02:38 AM	4d95e187d0045b63ff...	djm	1.1500			
Jun 22, 2026 11:05:...	Jun 22, 2026 11:05:24 PM	302459b94a66a5f16...	djm	1.0		64-bit	Windows11
Jun 23, 2026 2:02:...	Jun 23, 2026 2:02:38 AM	4d95e187d0045b63ff...	djm	1.1500			

Findings

The Windows 11 endpoint (djm) was successfully onboarded and actively reporting telemetry to Microsoft Defender.

Process Execution Telemetry

The DeviceProcessEvents table was reviewed to validate process execution monitoring.

Figure 5 – Process Execution Activity Identified Through Advanced Hunting

Advanced hunting

Help resources Query resources report Schema reference Page preferences

New query* +

Schema

Alerts & behaviors

- AlertEvidence
- AlertInfo
- BehaviorEntities
- BehaviorInfo

Devices

- DeviceEvents
- DeviceFileCertificateInfo
- DeviceFileEvents
- DeviceImageLoadEvents
- DeviceInfo
- DeviceLogonEvents
- DeviceNetworkEvents
- DeviceNetworkInfo
- DeviceProcessEvents
- DeviceRegistryEvents
- Defender Vulnerability Management
- DeviceBaselineComplianceAss...
- DeviceBaselineComplianceAss...
- DeviceBaselineCompliancePro...
- DeviceTvmBrowserExtensions
- DeviceTvmBrowserExtensionsKB

Query

1 DeviceProcessEvents

2 take 18

Results Query history Getting started

10 items Search 00:00:397 Low

Filters: Add filter

TimeGenerated	Timestamp	DeviceId	DeviceName	ActionType	FileName	FolderPath	SHA1
Jun 23, 2026 2:07:...	Jun 23, 2026 2:07:19 AM	4d95e187d0045b63ff...	djm	ProcessCreated	Usoclient.exe	C:\Windows\System32\...	587beee990eb
Jun 23, 2026 2:07:...	Jun 23, 2026 2:07:15 AM	4d95e187d0045b63ff...	djm	ProcessCreated	RuntimeBroker.exe	C:\Windows\System32\...	be41ebc53d7c
Jun 23, 2026 2:07:...	Jun 23, 2026 2:07:19 AM	4d95e187d0045b63ff...	djm	ProcessCreated	MoUsoCoreWorker.exe	C:\Windows\UUS\amd6...	506c3575bcb9
Jun 23, 2026 2:07:...	Jun 23, 2026 2:07:20 AM	4d95e187d0045b63ff...	djm	ProcessCreated	svchost.exe	C:\Windows\System32\...	230b124ae78a
Jun 23, 2026 2:07:...	Jun 23, 2026 2:07:20 AM	4d95e187d0045b63ff...	djm	ProcessCreated	backgroundTaskHost.exe	C:\Windows\System32\...	422ae532bcb5e
Jun 23, 2026 2:07:...	Jun 23, 2026 2:07:20 AM	4d95e187d0045b63ff...	djm	ProcessCreated	RuntimeBroker.exe	C:\Windows\System32\...	be41ebc53d7c
Jun 23, 2026 2:07:...	Jun 23, 2026 2:07:25 AM	4d95e187d0045b63ff...	djm	ProcessCreated	chrome.exe	C:\Program Files\Google...	7604f75db98a
Jun 23, 2026 2:07:...	Jun 23, 2026 2:07:25 AM	4d95e187d0045b63ff...	djm	ProcessCreated	chrome.exe	C:\Program Files\Google...	7604f75db98a
Jun 23, 2026 2:07:...	Jun 23, 2026 2:07:25 AM	4d95e187d0045b63ff...	djm	ProcessCreated	chrome.exe	C:\Program Files\Google...	7604f75db98a
Jun 23, 2026 2:07:...	Jun 23, 2026 2:07:25 AM	4d95e187d0045b63ff...	djm	ProcessCreated	chrome.exe	C:\Program Files\Google...	7604f75db98a

Findings

Process creation events were successfully collected, providing visibility into applications and system processes executing on the endpoint.

Network Telemetry Validation

The DeviceNetworkEvents table was queried to validate network connection monitoring.

Figure 6 – Network Activity Review Using DeviceNetworkEvents

Advanced hunting

Schema ▾

Alerts & behaviors

- AlertEvidence
- AlertInfo
- BehaviorEntities
- BehaviorInfo

Devices

- DeviceEvents
- DeviceFileCertificateInfo
- DeviceFileEvents
- DeviceImageLoadEvents
- DeviceInfo
- DeviceLogonEvents
- DeviceNetworkEvents
- DeviceNetworkInfo
- DeviceProcessEvents
- DeviceRegistryEvents

Defender Vulnerability Management

- DeviceBaselineComplianceAss...
- DeviceBaselineCompliancePro...
- DeviceTvmBrowserExtensions
- DeviceTvmBrowserExtensionsKB
- DeviceTvmCertificateInfo

Query

```
1 DeviceNetworkEvents
2 | take 10
```

Results

10 items

TimeGenerated	Timestamp	DeviceId	DeviceName	ActionType	RemoteIP	RemotePort	RemoteUri
Jun 23, 2026 2:35:...	Jun 23, 2026 2:35:34 AM	4d95e187d0045b63ff...	djm	SslConnectionInspected	20.42.73.24	443	
Jun 23, 2026 2:35:...	Jun 23, 2026 2:35:34 AM	4d95e187d0045b63ff...	djm	ConnectionAcknowledg...	20.42.73.24	443	
Jun 23, 2026 2:38:...	Jun 23, 2026 2:38:56 AM	4d95e187d0045b63ff...	djm	ConnectionSuccess	2603:1036:3000:60:d	443	login.microsoft
Jun 23, 2026 2:38:...	Jun 23, 2026 2:38:58 AM	4d95e187d0045b63ff...	djm	ConnectionAcknowledg...	2603:1036:3000:60:d	443	
Jun 23, 2026 2:39:...	Jun 23, 2026 2:39:29 AM	4d95e187d0045b63ff...	djm	ConnectionSuccess	52.182.143.214	443	us-v20.events
Jun 23, 2026 2:55:...	Jun 23, 2026 2:55:01 AM	4d95e187d0045b63ff...	djm	ConnectionSuccess	2603:1036:3000:148:2	443	login.microsoft
Jun 23, 2026 2:54:...	Jun 23, 2026 2:54:07 AM	4d95e187d0045b63ff...	djm	SslConnectionInspected	20.42.72.131	443	
Jun 23, 2026 2:54:...	Jun 23, 2026 2:54:07 AM	4d95e187d0045b63ff...	djm	ConnectionAcknowledg...	20.42.72.131	443	
Jun 23, 2026 2:56:...	Jun 23, 2026 2:56:04 AM	4d95e187d0045b63ff...	djm	ConnectionSuccess	193.138.111.30	443	cat.wargaming
Jun 23, 2026 2:56:...	Jun 23, 2026 2:56:04 AM	4d95e187d0045b63ff...	djm	ConnectionSuccess	127.0.0.1	11835	

Findings

Network telemetry included successful outbound connections, SSL/TLS activity, remote IP addresses, and internet communications.

Registry Telemetry Validation

The DeviceRegistryEvents table was reviewed to verify registry activity monitoring.

Figure 7 – Registry Activity Review Using DeviceRegistryEvents

Advanced hunting

The screenshot shows the Microsoft Defender Advanced Hunting interface. On the left is a 'Schema' pane with a search bar and a tree view of event categories: Alerts & behaviors, Devices, and Defender Vulnerability Management. The 'Query' pane on the right contains a query: `1 DeviceRegistryEvents` and `2 | take 10`. Below the query is a 'Results' pane showing 10 items. The results are displayed in a table with columns: TimeGenerated, Timestamp, DeviceId, DeviceName, ActionType, RegistryKey, RegistryValueType, and RegistryValueName. The table contains 10 rows of registry events.

TimeGenerated	Timestamp	DeviceId	DeviceName	ActionType	RegistryKey	RegistryValueType	RegistryValueName
Jun 23, 2026 2:03:23 AM	Jun 23, 2026 2:03:23 AM	4d95e187d0045b63ff...	djm	RegistryValueSet	HKEY_LOCAL_MACHINE...	Dword	DelayedAutostart
Jun 23, 2026 2:03:23 AM	Jun 23, 2026 2:03:23 AM	4d95e187d0045b63ff...	djm	RegistryValueSet	HKEY_LOCAL_MACHINE...	Binary	Configuration
Jun 23, 2026 2:03:23 AM	Jun 23, 2026 2:03:23 AM	4d95e187d0045b63ff...	djm	RegistryKeyCreated	HKEY_LOCAL_MACHINE...	None	
Jun 23, 2026 2:03:23 AM	Jun 23, 2026 2:03:23 AM	4d95e187d0045b63ff...	djm	RegistryKeyCreated	HKEY_LOCAL_MACHINE...	None	
Jun 23, 2026 2:03:24 AM	Jun 23, 2026 2:03:24 AM	4d95e187d0045b63ff...	djm	RegistryKeyCreated	HKEY_LOCAL_MACHINE...	None	
Jun 23, 2026 2:03:24 AM	Jun 23, 2026 2:03:24 AM	4d95e187d0045b63ff...	djm	RegistryValueSet	HKEY_LOCAL_MACHINE...	Binary	Blob
Jun 23, 2026 2:03:45 AM	Jun 23, 2026 2:03:45 AM	4d95e187d0045b63ff...	djm	RegistryValueSet	HKEY_LOCAL_MACHINE...	Binary	Blob
Jun 23, 2026 2:03:50 AM	Jun 23, 2026 2:03:50 AM	4d95e187d0045b63ff...	djm	RegistryValueSet	HKEY_LOCAL_MACHINE...	String	LastUpdate
Jun 23, 2026 2:05:36 AM	Jun 23, 2026 2:05:36 AM	4d95e187d0045b63ff...	djm	RegistryValueSet	HKEY_LOCAL_MACHINE...	Qword	KernelIntegrityCh...
Jun 23, 2026 2:14:47 AM	Jun 23, 2026 2:14:47 AM	4d95e187d0045b63ff...	djm	RegistryValueSet	HKEY_LOCAL_MACHINE...	Dword	TamperProtection

Findings

Registry key creation and registry value modification events were successfully captured and available for investigation.

Threat Hunt #1 – PowerShell Activity Analysis

Objective

Identify PowerShell execution activity on monitored endpoints.

Query

DeviceProcessEvents

| where FileName =~ "powershell.exe"

| project Timestamp, DeviceName, FileName, ProcessCommandLine

| take 50

Figure 8 – PowerShell Activity Identified Through Advanced Hunting

Advanced hunting

The screenshot shows the Microsoft Defender Advanced Hunting interface. On the left is a 'Schema' pane with a search bar and a tree view of data sources. The main area is titled 'Query' and contains a KQL query:

```
1 DeviceProcessEvents
2 | where FileName =~ "powershell.exe"
3 | project Timestamp, DeviceName, FileName, ProcessCommandLine
4 | take 50
```

Below the query editor is a 'Results' pane showing 15 items. It includes a table with columns: Timestamp, DeviceName, FileName, and ProcessCommandLine. The table lists several PowerShell execution events from June 23, 2026, on a device named 'djm'.

Timestamp	DeviceName	FileName	ProcessCommandLine
Jun 23, 2026 2:29:...	djm	powershell.exe	powershell.exe
Jun 23, 2026 2:30:...	djm	powershell.exe	"powershell.exe" -NoExit...
Jun 23, 2026 2:30:...	djm	powershell.exe	powershell.exe
Jun 23, 2026 2:52:...	djm	powershell.exe	powershell.exe
Jun 23, 2026 2:56:...	djm	powershell.exe	powershell.exe -NoExit ...
Jun 23, 2026 3:57:...	djm	powershell.exe	powershell.exe -Executi...
Jun 23, 2026 4:27:...	djm	powershell.exe	powershell.exe -Executi...
Jun 23, 2026 4:34:...	djm	powershell.exe	powershell.exe -Executi...
Jun 23, 2026 5:37:...	djm	powershell.exe	powershell.exe -Executi...
Jun 23, 2026 3:36:...	djm	powershell.exe	powershell.exe -Executi...
Jun 23, 2026 2:31:...	djm	powershell.exe	"powershell.exe" -NoExit...

Findings

Multiple PowerShell execution events were identified on the Windows 11 endpoint. Command-line arguments and execution timestamps were available for investigation.

MITRE ATT&CK Mapping

- T1059.001 – PowerShell

Threat Hunt #2 – Reconnaissance Activity Analysis

Objective

Identify command-line tools commonly used for system and network discovery.

Query

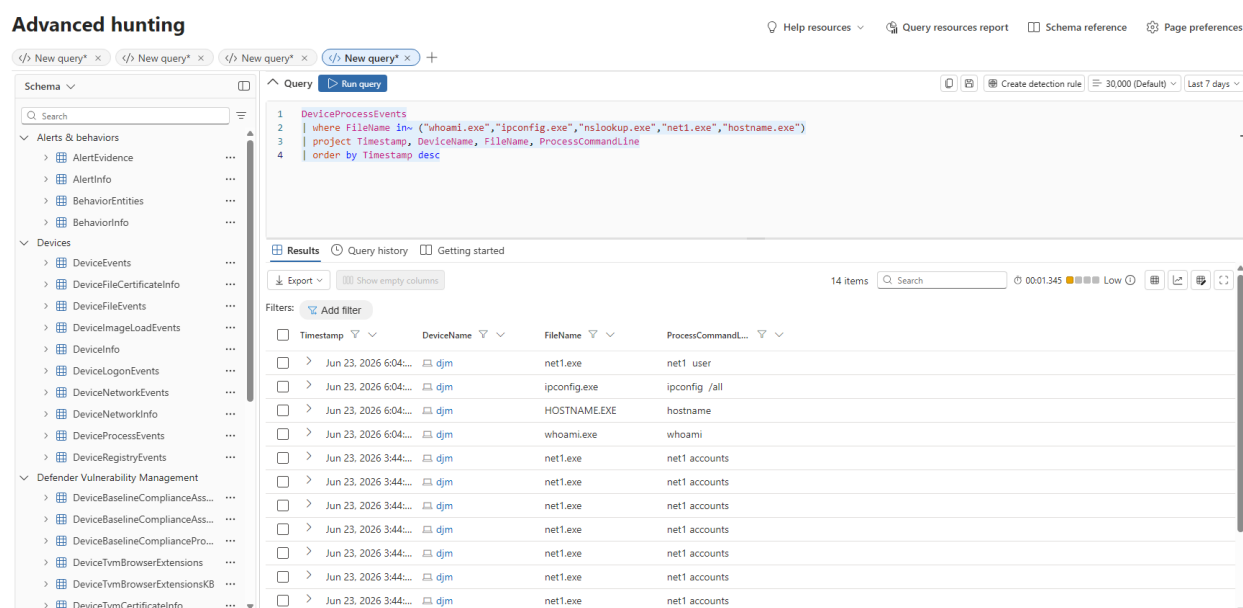
DeviceProcessEvents

| where FileName in~ ("whoami.exe", "ipconfig.exe", "nslookup.exe", "net1.exe", "hostname.exe")

| project Timestamp, DeviceName, FileName, ProcessCommandLine

| order by Timestamp desc

Figure 9 – Reconnaissance Activity Identified Through Process Analysis



Findings

Reconnaissance activity was identified through execution of discovery commands including whoami, hostname, ipconfig, and net user. These commands are commonly used to gather information about a target system.

MITRE ATT&CK Mapping

- T1082 – System Information Discovery
- T1033 – System Owner/User Discovery
- T1016 – System Network Configuration Discovery
- T1087 – Account Discovery

Threat Hunt #3 – Persistence Analysis

Objective

Identify registry modifications that may indicate persistence mechanisms or system configuration changes.

Query

DeviceRegistryEvents

| where ActionType in ("RegistryKeyCreated","RegistryValueSet")

| project Timestamp, DeviceName, ActionType, RegistryKey, RegistryValueName

| take 50

Figure 10 – Registry Modifications Identified During Persistence Hunt

Advanced hunting

Help resources Query resources report Schema reference Page preferences

New query* x New query* x New query* x New query* x +

Schema Search

Alerts & behaviors

- AlertEvidence
- AlertInfo
- BehaviorEntities
- BehaviorInfo

Devices

- DeviceEvents
- DeviceFileCertificateInfo
- DeviceFileEvents
- DeviceImageLoadEvents
- DeviceInfo
- DeviceLogonEvents
- DeviceNetworkEvents
- DeviceNetworkInfo
- DeviceProcessEvents
- DeviceRegistryEvents
- Defender Vulnerability Management
- DeviceBaselineComplianceAss...
- DeviceBaselineComplianceAss...
- DeviceBaselineCompliancePro...
- DeviceTvmBrowserExtensions
- DeviceTvmBrowserExtensionsKB
- DeviceTvmCertificateInfo

Query

```
1 DeviceRegistryEvents
2 | where ActionType in ("RegistryKeyCreated","RegistryValueSet")
3 | project Timestamp, DeviceName, ActionType, RegistryKey, RegistryValueName
4 | take 50
5
```

Results Query history Getting started

Export Show empty columns 25 items Search 00:00.853 Low

Filters: Add filter

Timestamp	DeviceName	ActionType	RegistryKey	RegistryValueName
Jun 23, 2026 2:14:...	djm	RegistryValueSet	HKEY_LOCAL_MACHINE...	TamperProtection
Jun 23, 2026 3:54:...	djm	RegistryValueSet	HKEY_LOCAL_MACHINE...	Health
Jun 23, 2026 4:09:...	djm	RegistryKeyCreated	HKEY_LOCAL_MACHINE...	
Jun 23, 2026 4:10:...	djm	RegistryKeyCreated	HKEY_LOCAL_MACHINE...	
Jun 23, 2026 4:10:...	djm	RegistryKeyCreated	HKEY_LOCAL_MACHINE...	
Jun 23, 2026 4:10:...	djm	RegistryKeyCreated	HKEY_LOCAL_MACHINE...	
Jun 23, 2026 4:10:...	djm	RegistryValueSet	HKEY_LOCAL_MACHINE...	Dhcpv6DomainSearchList
Jun 23, 2026 4:10:...	djm	RegistryValueSet	HKEY_LOCAL_MACHINE...	Dhcpv6Domain
Jun 23, 2026 4:10:...	djm	RegistryValueSet	HKEY_LOCAL_MACHINE...	Dhcpv6DNSServers
Jun 23, 2026 2:10:...	djm	RegistryValueSet	HKEY_LOCAL_MACHINE...	MDEAadTenantIdMappi...
Jun 23, 2026 2:53:...	djm	RegistryValueSet	HKEY_LOCAL_MACHINE...	Health

Findings

Registry modifications included key creation and value updates associated with operating system and security-related settings. Registry telemetry can provide indicators of persistence and configuration changes.

MITRE ATT&CK Mapping

- T1547 – Boot or Logon Autostart Execution
- T1112 – Modify Registry

Detection Engineering

Following the completion of threat hunting activities, custom detection logic was developed to automate monitoring of PowerShell activity.

Figure 11 – Custom Detection Rule Creation for PowerShell Activity

Custom detection

- General**
- Alert settings
- Automated actions
- Review and create

General

Proactively monitor and react to threats by creating custom detections that search through your logs.

Detection name *

Rule Description

Enter a description for your detection

Rule query *

```
DeviceProcessEvents  
| where FileName =~ "powershell.exe"  
| project Timestamp, DeviceName, FileName, ProcessCommandLine
```

 [Edit query](#)

Frequency * ⓘ

Continuous (NRT) ▼

RECOMMENDATION

This query looks well-optimized to run in near real-time. We have preselected the [Continuous \(NRT\)](#) frequency for you.

Lookback ⓘ

Near real-time rules run continuously as events are collected and processed, so there is no lookback period.

Severity *

Select severity ▼

[Next](#)

Figure 12 – PowerShell Detection Rule Configuration

Custom detection

General

Alert settings

Automated actions

Review and create

General

Proactively monitor and react to threats by creating custom detections that search through your logs.

Detection name *

Suspicious PowerShell Activity Detection

Rule Description

Detects PowerShell execution activity identified during threat hunting operations using Microsoft Defender Advanced Hunting telemetry.

Rule query *

```
DeviceProcessEvents
| where FileName -- "powershell.exe"
| project Timestamp, DeviceName, FileName, ProcessCommandLine
```

Edit query

Frequency * ⓘ

Continuous (NRT)

RECOMMENDATION
This query looks well-optimized to run in near real-time. We have preselected the [Continuous \(NRT\)](#) frequency for you.

Lookback ⓘ

Near real-time rules run continuously as events are collected and processed, so there is no lookback period.

Severity *

Medium

Next

Detection Logic

The detection rule was designed to identify PowerShell execution events and generate alerts for analyst review.

Severity

Medium

Frequency

Continuous (Near Real – Time)

Alert Enrichment and Correlation

Alert enrichment settings were configured to improve analyst context and streamline incident investigation.

Figure 13 – Alert Configuration and Enrichment Settings

Alert settings

Define the alert enrichment details.

 Explore query and results

Alert details *

Alert title

Suspicious PowerShell Activity Detected

Description *

PowerShell execution activity was detected on a monitored endpoint. Review the command line, initiating process, and associated user activity to determine whether the execution was authorized.

Custom details

Entity mapping *

Identified entities determine how alerts are grouped into incidents. Make sure to carefully review the entities to ensure a high quality of incident correlation. [Read more](#)

Findings

Alert metadata and contextual information were configured to improve investigation efficiency and incident triage processes.

Entity Mapping

Entity mapping was configured to associate detected activity with impacted devices.

Figure 14 – Entity Mapping Configuration for Device Attribution

Impacted Assets * ⓘ

Entity

Device

Identifier * ⓘ

HostName

Column * ⓘ

DeviceName

+ Add assets

Related Evidence ⓘ

+ Add entities

Findings

DeviceName was mapped to the HostName identifier, enabling improved alert correlation and asset attribution.

Findings and Recommendations

Findings

- Microsoft Sentinel successfully integrated with Microsoft Defender XDR.
- Endpoint telemetry was successfully collected and analyzed.
- Threat hunting activities identified PowerShell execution, reconnaissance commands, and registry modifications.
- Custom detection logic was successfully developed and configured.
- Alert enrichment and entity mapping capabilities were validated.

Recommendations

- Expand detection coverage to include encoded PowerShell commands and suspicious parent-child process relationships.
- Develop additional hunting queries focused on credential access and lateral movement.

- Create automated response workflows using Microsoft Sentinel automation rules.
- Implement scheduled threat hunting exercises to improve detection coverage.

Conclusion

This project demonstrated the successful deployment and utilization of Microsoft Sentinel and Microsoft Defender XDR for threat hunting and detection engineering activities. Endpoint telemetry was collected, analyzed, and leveraged to identify PowerShell execution, reconnaissance activity, and registry modifications. Detection engineering techniques were used to convert hunting logic into automated monitoring capabilities. The project provided hands-on experience with SIEM operations, threat hunting, incident investigation, and MITRE ATT&CK-aligned analysis, reflecting responsibilities commonly performed by Security Operations Center (SOC) analysts.

References

Microsoft. (2025). *Microsoft Sentinel documentation*.

Microsoft. (2025). *Microsoft Defender XDR documentation*.

MITRE. (2025). *MITRE ATT&CK Framework*.